

KONECRANES BOXHUNTER SPECIFICATION



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1. GENERAL

KONECRANES BOXHUNTER crane is capable of lifting and transporting ISO containers with a telescopic spreader. BOXHUNTER rubber-tired gantry crane is a mobile 8 wheeler gantry crane that is designed for handling of standard ISO containers (20-40ft) within the stacking areas of a container terminal.

It is designed to perform common lifting, travelling, and lowering operations, within the limits specified by the equipment's duty group (see section Duty class) on fenced site area.

2. OPERATION

The crane performs the following operations on a duty cycle basis:

The crane picks up a container from a stack or ground level and deposits it on the ground or on a truck chassis or similar. It also reverses this process.

The crane travels through narrow lanes separating container stacks either with or without load suspending from the spreader.

To speed the duty cycle A container can be hoisted or lowered at the same time the trolley is traversed, and

The gantry can be travelled at the same time the trolley is traversed.

The individual control of the gantry drive motors permits the crane to maintain or change its travel direction.

The crane is also capable of changing lanes or working areas, without load under spreader and with reduced speed, as the wheels are pivoted through 90 degrees around their central vertical axis to the cross gantry mode position.

When the wheels are aligned for the gantry cross travel motion, the trolley traverse motion is disabled.

2.1. OPERATOR CABIN AND INTERFACE



The crane is operated by a single operator from an enclosed cabin, which is located on access side sill beam. Cabin is in an optimal position for loading and unloading containers in truck lane with visibility to operating area. BOXHUNTER is controlled with Controller console instead of traditional joystick controllers.

Cameras and camera monitor are used for positioning the spreader in the container stack and while gantry travelling. In Camera monitor spreader and motion (hoist/trolley/ gantry travel/ general on top of stack) view is visually displayed. Standard Lift truck cabin is used.

The cabin structure protects the operator from falling objects. There is beam frame in front of the cabin for cabin safety

Cabin is air-conditioned and equipped with seat for providing comfortable working environment for the operator.

Also an operator's display Panel is located in the Operator's Cabin. The panel gives general operational information and assists the operators and displays alarm messages.

Alarm horn and loud speaker are available for communication with a truck driver.

2.2. DOCK INTERFACE AND WHEEL LOADS

The gantry travel runway may be of asphalt, concrete, or other material suitable for the wheel loads ground pressures specified in Annex 4.

The maximum gradient of the gantry travel runway shall not exceed 2% in any direction.

3. DESIGN STANDARDS

General designs standards are:

Structure:	FEM, EN
Mechanisms:	FEM, ISO, EN
Electrical:	IEC
General safety, walkways and service platforms:	FEM, EN

3.1. GENERAL DESIGN CRITERIA

The crane structure is designed according to EN 15011 U7/Q3 classification (FEM A7/U7/Q2)

FEM Section 1 (1987):

Class of utilization	U7 (2 million cycles)
Spectrum class	Q2
Appliance group	A7

Motors, gearboxes and shafting of the hoist, trolley traverse and gantry travel machinery have the adequate power and rating to provide the accelerating and decelerating forces required to enable the normal operating cycle to be worked continuously when handling the rated load.

Ambient conditions for the crane is:

Ambient temperature	-0°C ... +45°C
Relative humidity	up to 100%

Design loads of the crane are:

Rated load under spreader	40.6 ton (40 LT)
Load for fatigue design	26.0 t concentric load, including impact effects (This load is assumed to be equivalent to the actual spectrum of various weight containers.)

The impact factor is 1.15 for calculation of structures and mechanisms.

The crane is designed to handle eccentrically loaded containers. The maximum eccentricity of the container is 10% in either direction (1.22 m on container longitudinal and 0.24 m on transversal axis).

Maximum operating wind speed	22 m/s
Maximum out-of-service wind speed	44 m/s

The classification of the mechanisms according to FEM Section 1 (1987) is as follows:

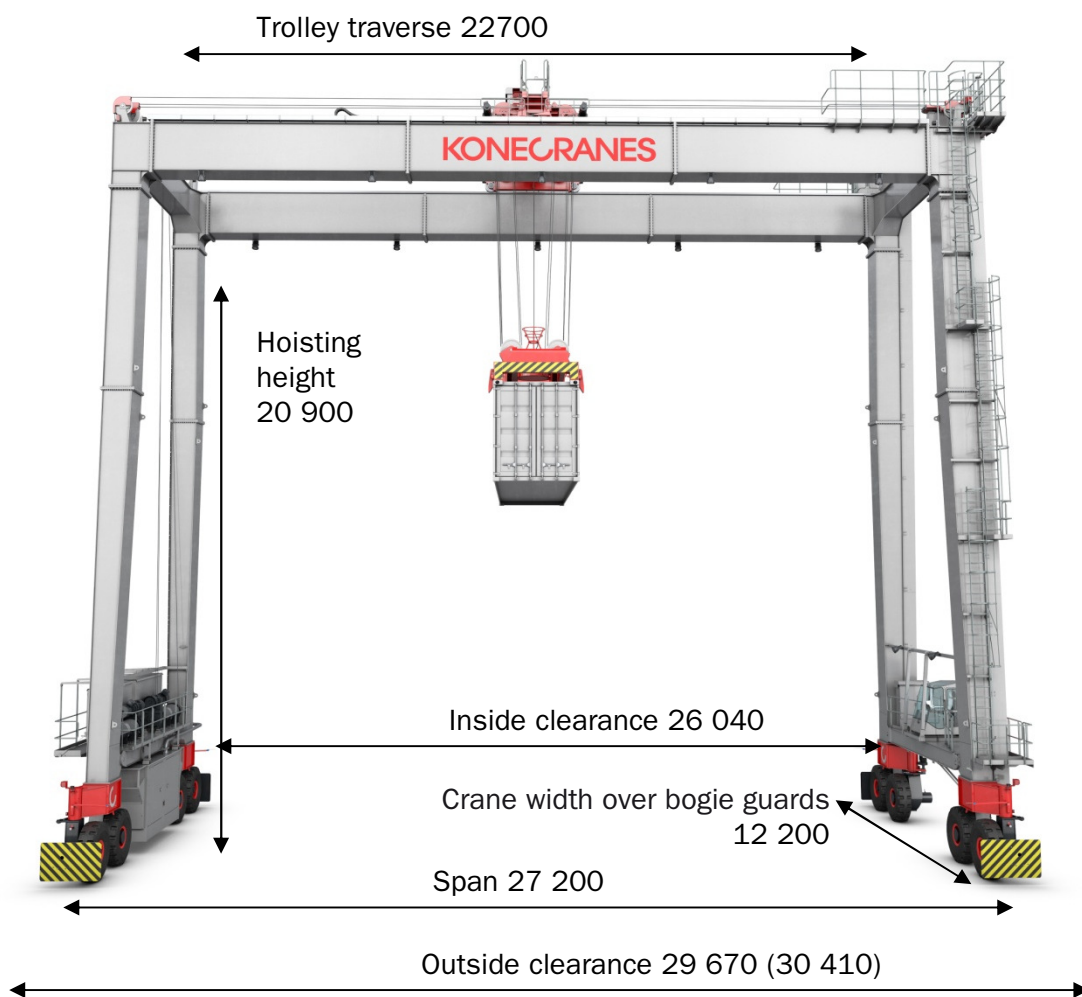
Mechanism	Min. life	Class of Operation	Load Spectrum	Classification
Hoist	25,000 h	T7	L2	M7
Trolley Traverse	25,000 h	T7	L2	M7
Gantry Travel	6300 h	T5	L2	M5

3.2. DUTY CYCLE

3.3. PRINCIPAL DIMENSIONS

Maximum lifting capacity 40,6t /40LT

Maximum dimensions available with 1 over 6 and 7 + truck lane:



	Boxhunter - Dimensions with max 1 over 6 and max 7+lane	8W	16W
1	Lifting capacity	40.6 t	40.6 t
2	Max. Lifting height (1over6) / Max span (7+truck lane)	20900 / 27200	20900 / 27200
3	Extension from leg centerline at diesel (with foldable platform open) / crane access side	1080 (1820) / 1160	1080 (1820) / 1160
4	Outside/inside clearance at bogie and e-cubicle row level	29670 (30410) / 26040	29440 (30180) / 25700
5	Crane width over bogie guards / wheel spacing in bogie	12200 / 2000	11910 / 2100

4. HOISTING SYSTEM

4.1. HOISTING MACHINERY



The hoist machinery consists of two electrically synchronized machinery units mounted on Genset side sill beam.

Each hoist machinery unit consists of:

- A single layer rope drum, one end of the drum supported by a spherical roller bearing, the other end supported by low speed shaft of the gear reducer
- Flexible drum coupling manufactured by KONECRANES
- Foot mounted totally enclosed gear reducer
- Disc brake on high speed shaft extension of the gear reducer
- Flange mounted AC motor, connected to the high speed shaft of the gear reducer by a gear coupling

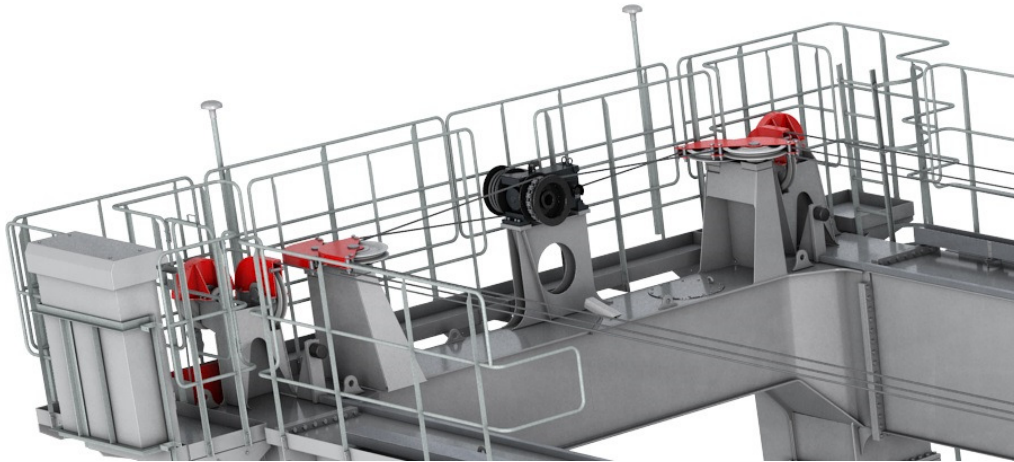
The hoist rope drums are of fabricated welded construction with machined grooves for single layer winding of the hoist ropes. Flanges are provided at both ends.

Drum and pulley dimensioning is found in Annex 1.

Each rope is secured to the drum by clamping devices.

Secondary torque of the hoisting machinery is reduced by counterweight system, which is dimensioned to compensate spreader weight. See more in section 4.4 Counterweight system.

4.2. SKEW MACHINERY



Skewing is rotation of the spreader around its vertical axis. The skew range is $\pm 5^\circ$.

Skew of the spreader is achieved by shortening one main hoist rope and lengthening the other hoist rope with one skew machinery located at the end of main girder.

Skew machinery unit consists of:

- Flange mounted AC motor direct connected to the high speed shaft of the gear reducer
- Motor mounted disc brake
- Fully enclosed hollow shaft gear reducer

4.3. ROPE SPECIFICATION AND REEVING

The hoist wire ropes are of 6x36 IWRC Warrington-Seale construction.

Metric size wire ropes are used. The nominal diameter of the hoist wire rope is 18 mm.

A minimum safety factor of the hoist wire ropes is over 5 against ultimate breaking strength with rated load under spreader.

The ultimate breaking strength of the wire ropes is 1960 N/mm².

Counter Weight ropes are similar to hoisting ropes, and the safety factor of the ropes is 25 against ultimate breaking point.

Rope reeving is found in Annex 3.

4.4. COUNTERWEIGHT SYSTEM

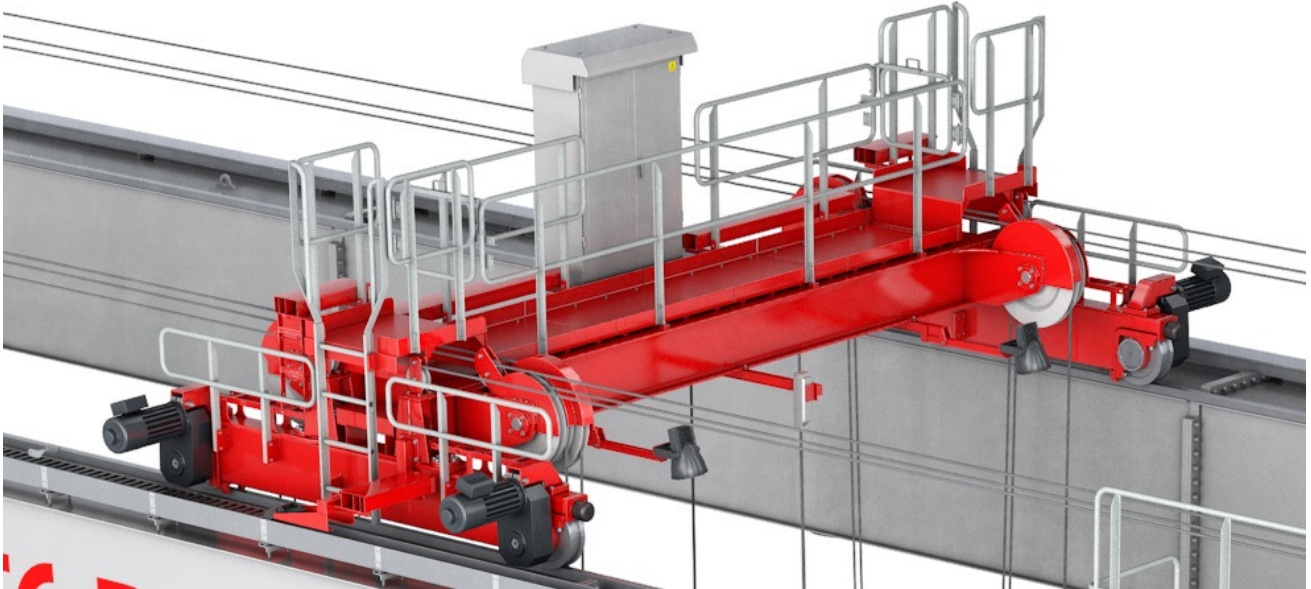


Counter Weight system is used as energy storage and it compensates partly the weight of empty spreader. With Counterweight system required energy consumption is reduced.

Each hoisting machinery unit has its own counterweight unit, which weigh approximately 3 tons each. Wire ropes are installed in the same drums as main hoisting ropes.

Counterweights run along the access side leg structure using elevator like guide rail (ISO 7465) and counterweights have emergency stop claws. Counterweights are covered for safety.

5. TROLLEY



The trolley is a structural frame mounted on four flanged steel wheels travelling on the rail of the main girder beams. Horizontal girder beams which supports rope pulleys are bolted on the end carriages.

Double flanged trolley wheels machined from cast iron with a rim hardness to match the rail properties are used. All four trolley wheels are driven.

Each trolley wheel has a drive machinery and each machinery unit consists of:

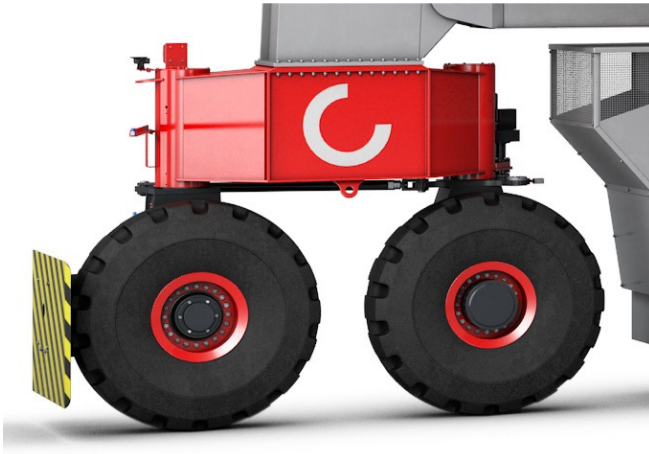
- Flange mounted AC motor direct connected to the high speed shaft of the gear reducer
- Motor mounted disc brake
- Fully enclosed hollow shaft gear reducer mounted on the rail wheel shaft by Konecranes.
- Flexible shock absorbing torque arm at primary end of the gear reducer ,

Rails of square type, 60 mm wide and 60 mm high, are welded on the upper flange of both main girder beams. The rails are located concentric over the inner webs of the main girder beams.

Stopper plates on main girder rails at both ends of the trolley runway and rubber buffers at both ends of the trolley are provided. This system is rated to safely stop a fully loaded trolley travelling at full speed with the power off. Safety catches are provided to prevent the trolley falling from the crane in the event of any loading. Drop blocks are provided to support the trolley in the event of an axle or bearing failure.

Trolley is equipped with 4 floodlights, camera system and electrical cubicle.

6. GANTRY TRAVEL AND BOGIES



The gantry travel system has two operational modes;

In NORMAL operational mode the gantry wheels are parallel with the sill beams. The operator can steer the crane from the operator's cab. Based on the commands given by the operator the control system of the crane varies the speed of the gantry motors at one sill beam relative to the motors at the other sill beam in order to correct any misalignment of the crane.

In CROSS GANTRY mode the crane is capable of travelling 90 degrees to the normal operating direction as all wheels are pivoted through 90 degrees.

In PARKING mode the wheels at opposite corners are pivoted through 90 degrees to hold the crane against moving in high wind conditions.

The crane is supported by eight (8) pneumatic tired wheels, two wheels in line at each corner. One wheel per corner is driven. Driven wheel is mounted on the output hub of the gear reducer. The non-driven wheel is supported on anti-friction bearings.

The gear reducer of each gantry travel machinery unit is fully enclosed, oil bath lubricated and planetary type. Flange mounted AC motor is connected directly to the high speed shaft of the gear reducer. Each motor is provided with an integral disc brake, which is provided with means for manual release.

Each wheel support beam is connected to the lower end of the leg through bolted joint.

Each bogie is equipped with gantry travelling alarm device, runway camera and collision prevention devices against stacked containers and obstacles on runway.

7. SPREADER

The All Electric Single Lift Telescopic Reeved-In Spreader is suitable for handling 1 x 20ft, 1 x 40ft or 1 x 45ft ISO containers and designed to be fitted to a rubber tired gantry crane. The spreader is fitted with built-in rope sheaves for connection to the crane rope system.

Spreader frame is equipped with rope pulleys, pulley pockets and easily removable pulley pins for maintenance and spreader change.

There are electrically operated twistlocks at each corner of the spreader that are used for attaching into the top corner castings on a container. The twistlocks are operated by two separate electric motors and gearboxes.

The spreader is fitted with four heavy construction fixed gather guides fitted to the longitudinal side corners of the spreader.

The spreader is all electric, with all functions controlled from the operator's console within the crane.

A stainless steel junction box is fitted to the center of the spreader with elastic suspension elements.

The spreader is fitted with 4 lifting lugs each of 10 tonnes capacity at each corner of the spreader.

The spreader is fitted with a heavy steel construction cable basket (Konecranes type) at top of center section for gathering the connecting cable.

The spreader is fitted with a camera system having 4 cameras at twistlock positions and adjustable mounting brackets. The technical specification of the telescopic spreader is found in Annex 6.

8. POWER SUPPLY OPTIONS

There are three alternatives for RTG main power supply, two speed diesel alternator set, Cable reel and Conductor rail (Bus Bar).

8.1. DIESEL GENSET



The main power supply for all crane functions and accessories is provided by a diesel alternator set containing a diesel engine powered AC alternator mounted onto a common base. The diesel alternator set is located in a weatherproof sound insulated enclosure suspended from the sill beam.

The output of the diesel alternator set is sufficient to start and accelerate simultaneously both the hoist and trolley traverse motion as well as the trolley traverse and gantry travel motion to full speed when handling the rated maximum load, and to supply all other services in use (lighting, heating, air-conditioning, communications, etc.).

The diesel engine is equipped with all necessary accessories needed for its proper operation and operates with standard diesel fuel oil. A diesel fuel system water separator is provided.

The diesel engine is equipped with a battery operated electrical starting system.

Diesel Genset is equipped with two speed fuel saving system. In normal operation engine is governed to fixed full running speed and decreased to idle speed during non-operation time.

Exhaust pipe is lead to main girder level.

8.2. CABLE REEL

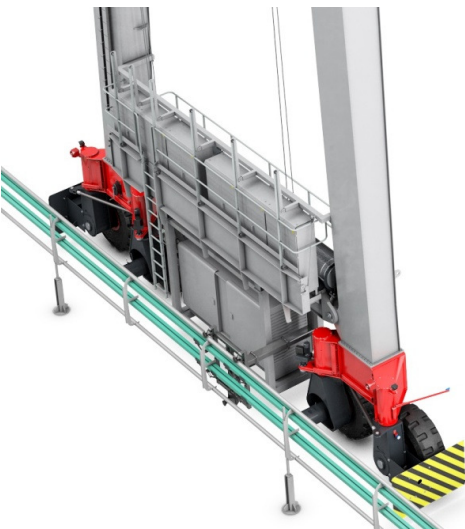


The crane is equipped with a cable reel and transformer to feed the energy for the main functions. The power supply for the crane will come from the Customer's power station. The power is fed from the yard to the crane by flexible cable collected automatically on a cable reel on the crane. The cable is attached to electricity feeding point with a plug. The feed point can be either at one end of the operating area or at the center. Supply voltage 1000V (Preferable), A fixed or portable auxiliary diesel generator is used to move the RTG between stacks

Transformer and cable reel control equipment are located in a weatherproof sound insulated enclosure under hoisting machinery sill beam.

Yard infrastructure including feeding cubicle, suitable sockets and friction rings for cable protection are provided by the customer.

8.3. CONDUCTOR RAIL (BUS BAR)

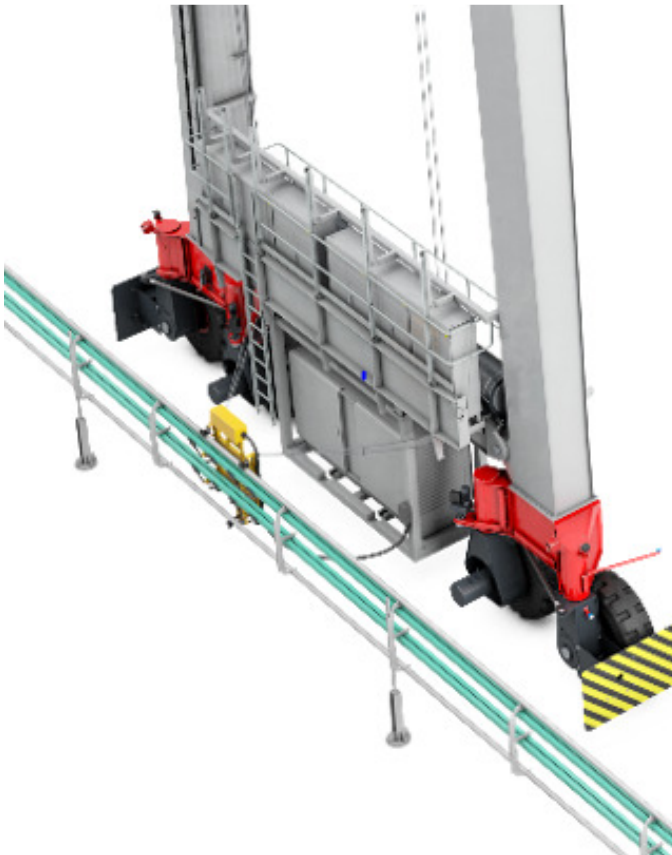


The crane is equipped with a telescopic arm with current collector trolley for semi-automatic connection of the crane to the conductor rails.

Necessary step-up or step-down devices for feeding the energy to the crane's electric system are provided. The power supply is from the Customer's power station via Customer's conductor bar system.

The power is supplied by a low electrified fence that provides the physical contact. 480V (Preferable), Depending on the voltage, a transformer may be required. A fixed or portable auxiliary diesel generator is used to move the RTG between stacks

8.4 PLUG-IN TROLLEY



The power is supplied by a low electrified Plug-in trolley that travels besides BXH. Supply voltage 480V (Preferable), depending on the voltage, a transformer may be required. A fixed or portable auxiliary diesel generator is used to move the RTG between stacks

The basic idea is to use an Plug-In System, where the power supply of the conductor rail system must be disconnected manually via a plug and socket connector. This manual connection can be easily realized by the crane operator as the driver's cabin has been brought down on the Boxhunter.

9. ELECTRICAL EQUIPMENT

9.1. CRANE CONTROL SYSTEM

AC squirrel cage motors are used for hoist, trolley traverse, gantry travel and skew. The motors are powered through inverters (frequency converters).

A stepless speed adjustment is achieved by changing squirrel cage motor's supply voltage frequency.

The control is PLC based and arranged to interface with operating, control, limiting and sensing devices through the necessary Input/Output modules. PLC also monitors logic and memory functions for diagnostic and adjustment purposes.

9.2. LIMIT SWITCHES AND INTERLOCKS

The crane is provided with limit switches and interlocks for safe operation.

The switches are so arranged that, if one of the contact signals in the unit is operated while travelling in the given direction, it will remain in that condition and will reset only when rotated in the opposite direction.

The emergency stop switches are hard wired to emergency stop circuit and also to PLC for monitoring.

All switches are dust-tight and watertight and suitable for heavy duty, severe impact and vibration in marine use.

9.3. OVERLOAD PROTECTION

Total four overload sensors are located between hoist wire ropes near skew machinery.

9.4. CRANE MONITORING SYSTEM

The Crane monitoring system is available through internet. The system monitors regular and irregular process and state information of the crane. The monitored data is divided into groups to improve crane service and management. The data groups are:

- Alarm and fault messages
- Counters
- PLC I/O
- Maintenance

9.5. ENCLOSURES

Fan ventilated electrical cubicles are at following locations:

Majority of the cubicles are located in the main cubicle row on the same sill beam as hoisting machinery.

Operator related electrical components are located in electrical cubicle on sill beam near operator's cabin.

Skew machinery related components are located in electrical cubicle on main girder near skew machinery.

Trolley related components are located in electrical cubicle on trolley.

No e-room is needed.

10. CRANE STRUCTURE

10.1. MAIN STRUCTURE

Main girders, legs and sill beams are box girders.

All exterior welds in the main structure are continuous seal welds. Intermittent welds are used only in some interior parts of the structures.

All load bearing welded plate structures are 8mm or thicker, except for internal stiffener angles of box girders. Main girders are fabricated in sections and sections are connected with bolted flanges. Also the connections between the legs and main girder beams and legs and bogie structures are of bolted type. Pin joints are used between the legs and sill beams.

10.2. LIGHT STRUCTURES

Walkways are used in hoisting machinery side, cabin access, Main girder end at the trolley parking position side and on trolley. Access to other areas is done with mobile equipment or similar.

Ladders are used for vertical passage.

Cables are located in cable trays on main girder for different voltage and cable categories. Cable trays are used also on trolley. Cable- T (Double L-profile) structures are used for protecting the cables on vertical parts of legs.

Cable channels are used from leg to bogies.

11. SURFACE TREATMENT

Crane surface treatment designed for atmospheric corrosivity category C4 according to ISO12944.
Maintenance durability is medium.

Surface treatment done according Annex 2:

12. DELIVERY CONCEPT

BOXHUNTER is a standard product with few options available. Product is configured.

Options available:

Lifting height:	1 over 5 or 1 over 6
Span	6 + trucklane or 7 + trucklane
Bogies:	8W or 16W
Power source alternatives:	diesel, busbar, plug-in trolley, cable reel
Autosteering	

12.1. MANUFACTURING

Manufacturing is centralized in a production hub located in Asia. With this high quality, focused and efficient serial production is achieved. All components are assembled into appropriate modules to minimized site assembly and enable standard container delivery.

Long lead time modules are stored at warehouse to minimize the delivery time.

12.2. TRANSPORTATION

All modules are transported in containers to erection site.



12.3. SITE WORKS

Local Site works can start right after the arrival of container deliveries. Site area of 70 x 100 meters is needed for erecting every two cranes. Mobile equipment such as fork lift, telescopic man lift is needed during the erection period and for lifting the BOXHUNTER 2x 100t or one 150t mobile crane is needed.

12.4. COMMISSIONING AND TESTING

Commissioning is done according to attached commissioning and testing procedure, Annex 5.

12.5. TRAINING

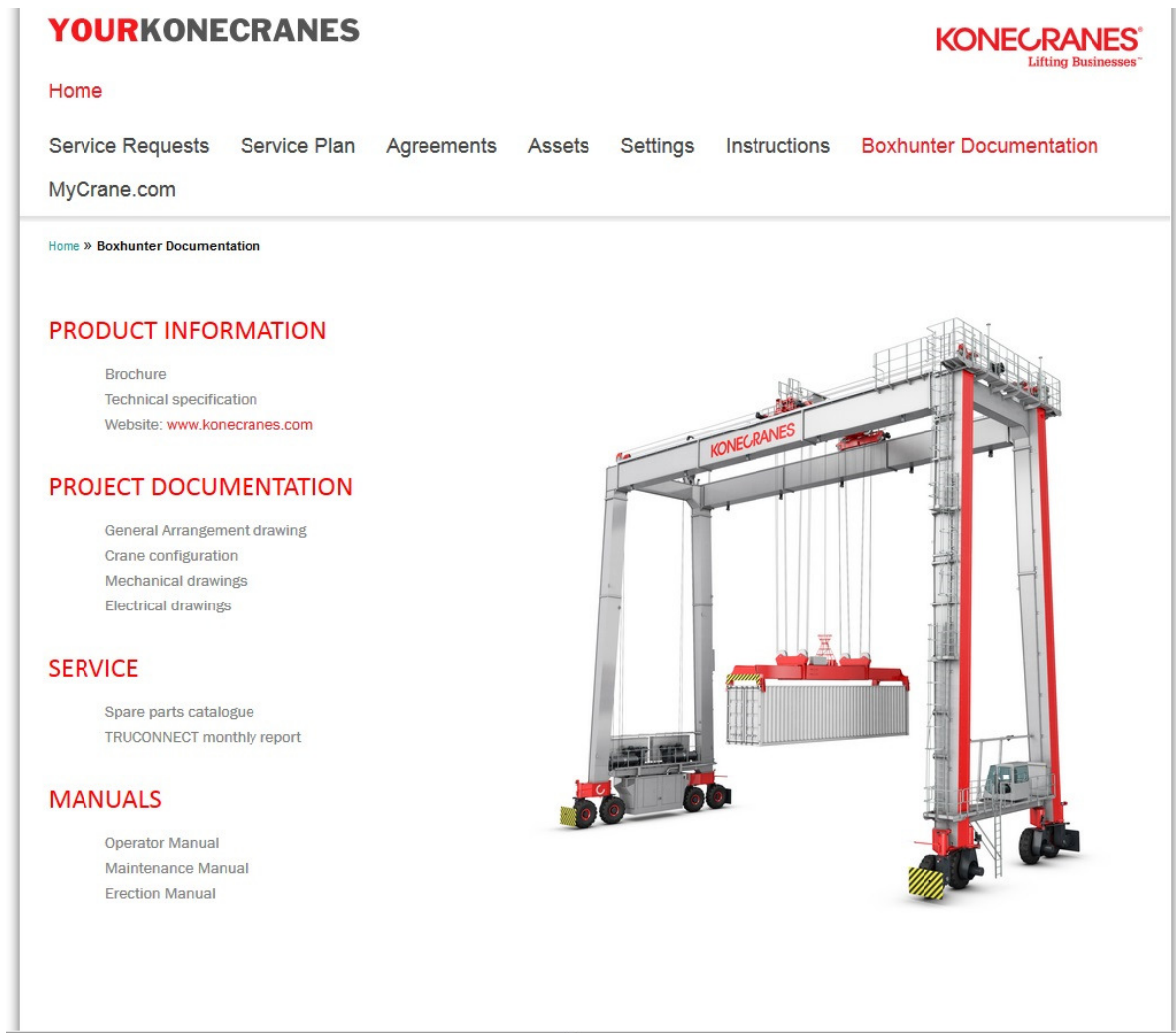
A comprehensive program is provided for both theoretical and practical training for the Customer's mechanical and electrical maintenance people and crane operators.

The training sessions are performed in a classroom arranged by the Customer and on the crane immediately after completion of the test runs.

The trainees are assumed to have a basic practical working knowledge on their respective trades.

See Annex 10 for basic training program

13. DOCUMENTATION



All documentation is available at our customer web portal:

<http://your.konecranes.com>

User rights with passwords are obtained from Konecranes responsible Sales Manager.

Documents available:

- Product information
- Project and delivery related documents, like Mechanical and Electrical drawings
- Service documents, like spare parts catalogue
- Operator and maintenance manuals

Language of documentation is English.

14. SERVICE CONCEPTS

As part of the product we have three service packages available:

- Annual inspection
- Quarterly inspection
- Preventive maintenance

And for Spare parts also three alternatives

- Basic, with minimum electrical components
- Standard, including some wear and tear parts
- Premium, more extensive and including investments parts

More details are found in the web portal:

<http://your.konecranes.com>

TRUCONNECT connection is included in crane deliveries.



15. ANNEXES

Annex 1

Crane parameters

Annex 2

Surface treatment instructions

Annex 3

Rope reeving

Annex 4

Wheel loads

Annex 5

Testing specification

Annex 6

Spreader

Annex 7

Tech specification 8W

Annex 8

Tech specification 16W

Annex 9

Supplier list

Annex 10

Training program



Annex 1 Crane Parameters

Dimensions with max. 1 over 5 and 1 over 6 max. 7+ truck lane, mm	8 wheels [16 wheels]
Lifting capacity 40.6 t	
Max. lifting height (1 over 5) / Max. span (7 + truck lane)	18 100 / 27200mm
Extension from leg centerline at diesel (with foldable platform open)/crane access side	1140 (1900)/ 1260mm
Outside/inside clearance at bogie and e-cubicle row level	29600 (30360) / 26 138mm
Crane width over bogie guards/wheel spacing in a bogie	12 198/ 2000mm
Speeds, max., m/min	
Hoist with empty spreader/35/40.6 ton load	52/26/23
Trolley traverse	70
Gantry travel with empty spreader	130
Accelerations	
Hoist from 0 to full speed with empty spreader/40.6 ton load	2/4 sec
Trolley from 0 to full speed with empty spreader/rated load:	5/5 sec
Gantry travel from 0 to full speed(no load, no wind, no slope)	8 sec
Deceleration times are same as acceleration times	
Trolley	4-wheel drive
Type of anti-sway system included	Electric
Type of micro-motions	Spreader side-shift, optional
Skew/trim angle, degrees	±5°
Bogies	4 wheels driven [8 wheels driven]
Tire size/pressure, bar	18.00 x 25/9.5 (D1673 mm), [14.00-24/9.5]
Wheel load, tons with 1 over 5 and 6+lane, max. load, no wind	27,5 [13.3]
Control System	
Drive	Konecranes
PLC	Siemens
BOXHUNTER operating system	
Spreader	Reeved-in spreader with integrated headblock
Cabin	Standard Konecranes lift truck cabin
Operating equipment in cabin	Konecranes graphical user interface, OP panel, controller
Video camera & laser scanning system	Konecranes
Crane Management System (CMS)	Konecranes
Default power supply	
Diesel alternator set	Volvo TAD1351GE, EU Stage 3, Max 351 kW
Main options	
DGPS Autosteering / Container Positioning System	With dual antenna technology for

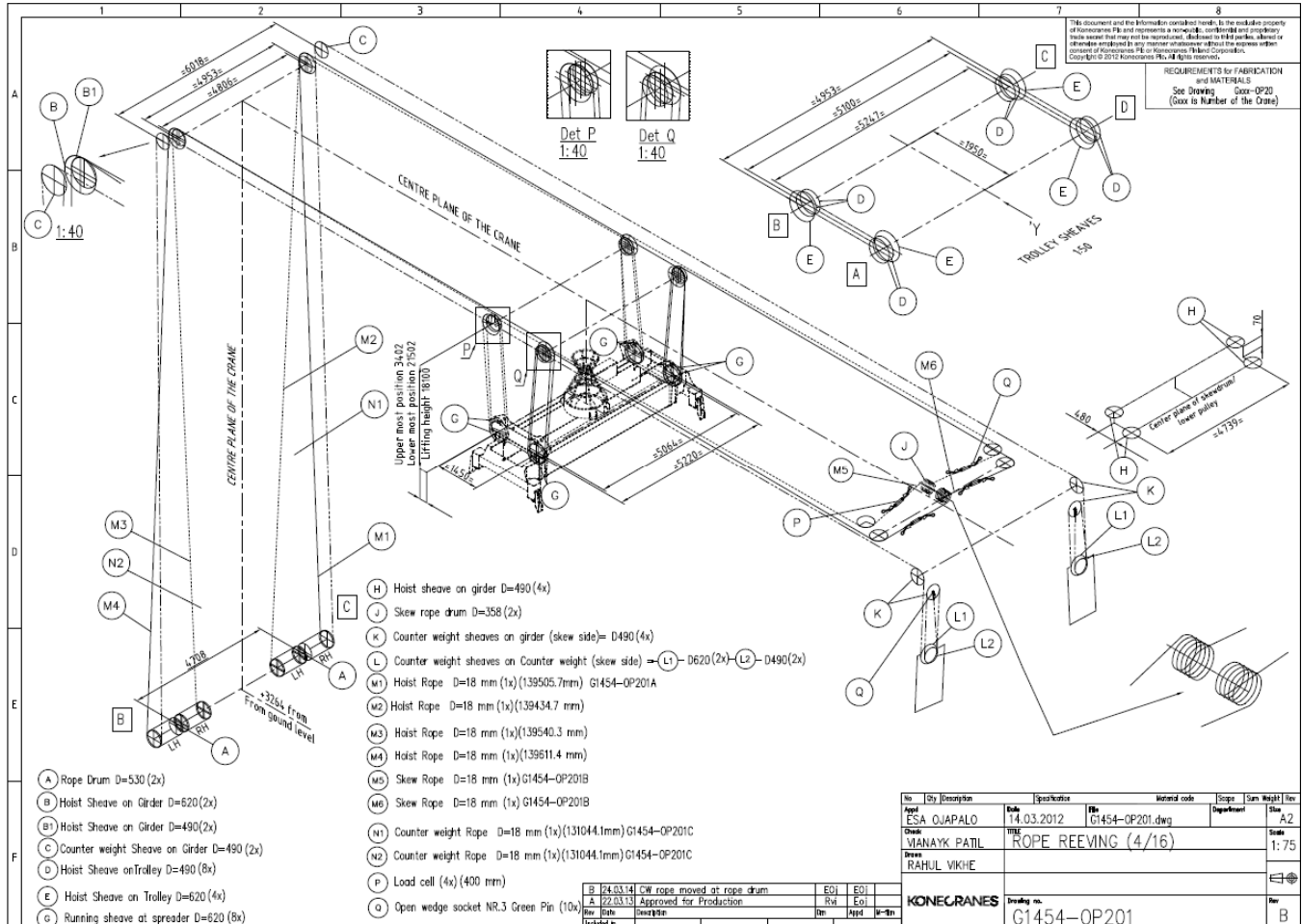
	accurate real-time heading info and rapid signal recovery
Diesel Fuel Saver System	Two-step, Stage II-III
Cable reel power supply instead of diesel engine/alternator	3-phase, 50/60 Hz, 1 kV
Busbar power supply instead of diesel engine/alternator	400-500V
Plug-in Trolley supply instead of diesel engine/alternator	400-500V
Other parameters	
Pitch diameter of the hoist rope drums	29,4 times the rope diameter
Pulley pitch diameter at hoisting machineries	28 times the rope diameter
The material of the rail	S355J0 (EN10025:1993), equal to Q355
The steel material of the main structure	S235J0 EN10025:1993 (or Q235)

Annex 2

See OP3 Surface treatment instruction

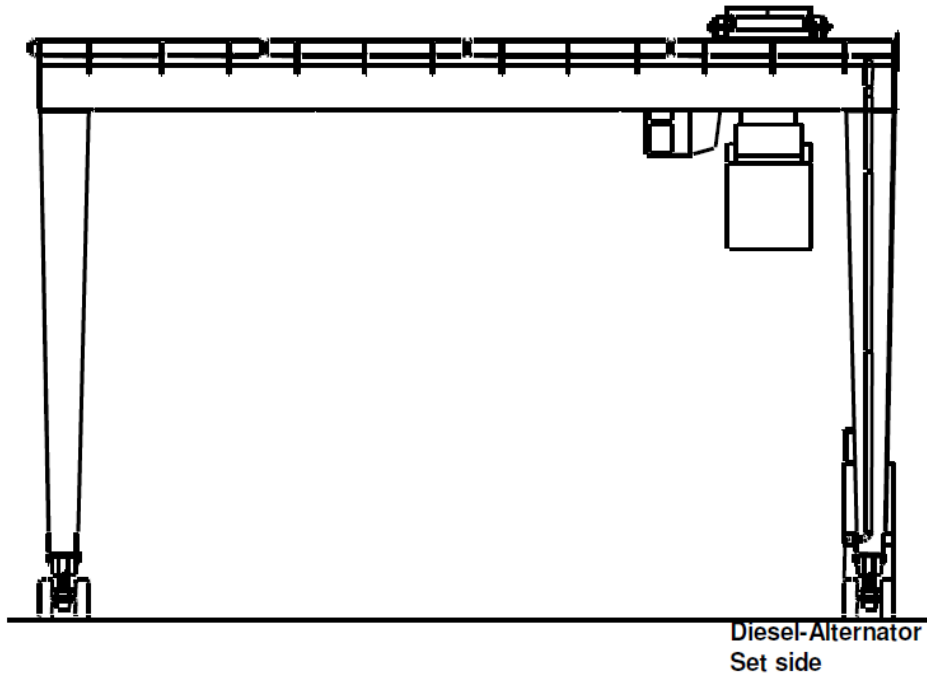
Annex 3 Rope reeving

See drawing G1454-OP201



Annex 4 Wheel loads

8W RTG
1 over 5_6+lane_40LT



Dead weight & Wheel loads at Diesel-Alternator set side

Dead weight: 127536
Container weight max: 40600
Container weight average: 26000
Number of wheels: 8

MAXIMUM WHEEL LOADS at Diesel-Alternator set side: (Tyre pressure 9.5 bar, 138 PSI)

Container Weight (kg)	Wheel load (ton)	Gross contact area cm ²	Gross ground pressure kg/cm ²
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Without dynamic loads, without wind and with container

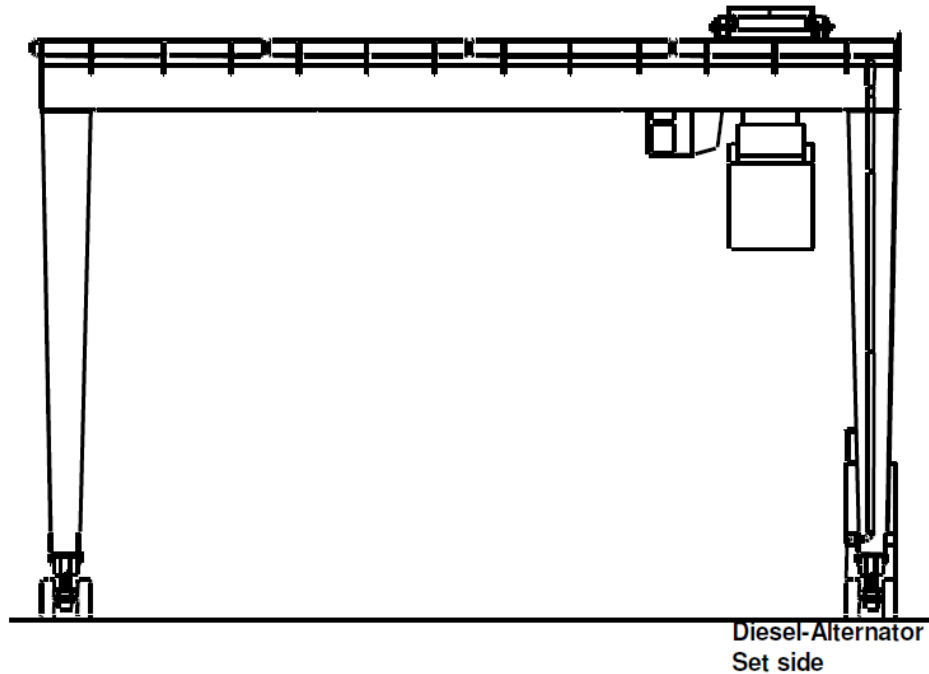
0	19,9	2189	9,1
26000	25,8	2773	9,3
40600	29,1	3102	9,4

Without dynamic loads and with operating wind of 20 m/s
and with container

0	24,8	2676	9,3
26000	30,7	3261	9,4
40600	33,9	3589	9,5

Annex 4 Wheel loads

8W RTG
1 over 5_7+lane_40LT



Dead weight & Wheel loads at Diesel-Alternator set side

Dead weight: 131852
Container weight max: 40600
Container weight average: 26000
Number of wheels: 8

MAXIMUM WHEEL LOADS at Diesel-Alternator set side: (Tyre pressure 9.5 bar, 138 PSI))

Container Weight (kg)	Wheel load (ton)	Gross contact area cm ²	Gross ground pressure kg/cm ²
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Without dynamic loads, without wind and with container

0	20,6	2251	9,1
26000	26,5	2843	9,3
40600	29,8	3176	9,4

Without dynamic loads and with operating wind of 20 m/s and with container

0	25,8	2773	9,3
26000	31,7	3365	9,4
40600	35,0	3697	9,5

Annex 5 Testing specification

Annex 6 Spreader

ALL ELECTRIC TELESCOPIC SPREADER WITH BUILT-IN ROPE SHEAVES

GENERAL DESCRIPTION

The All Electric Single Lift Telescopic Reeved-In Spreader is suitable for handling 1 x 20ft, 1x30ft, 1 x 40ft or 1 x 45ft ISO containers and designed to be fitted to a rubber tyred gantry crane. The spreader is fitted with built-in rope sheaves for connection to the crane rope system.

The spreader structure is an all welded steel assembly with a center section supporting the two telescopic arm assemblies. The telescopic arms are stationed using electrically operated stops at the 20ft,30ft,40ft and 45ft positions. The telescopic drive is operated by a single electric motor and gearbox.

There are electrically operated twistlocks at each corner of the spreader that are used for attaching into the top corner castings on a container. The twistlocks are operated by two separate electric motors and gearboxes.

The Spreader is fitted with LED indicator light panel having three lights; red, green and yellow. The panel is fitted on top of spreader center section to be visible to crane operator.

The spreader is fitted with four heavy construction fixed gather guides fitted to the longitudinal side corners of the spreader.

The spreader is all electric, with all functions controlled from the operator's console within the crane.

A stainless steel junction box is fitted to the center of the spreader with elastic suspension elements.

The spreader is fitted with 4 lifting lugs each of 10 tonnes capacity at each corner of the spreader.

The spreader is fitted with a heavy steel construction cable basket (Konecranes type) at top of center section for gathering the connecting cable. The spreader is fitted with a camera system having 4 cameras at twistlock positions and adjustable mounting brackets.

Main Particulars:

Lifting Capacity:	40.0 Long Tons (SWL), (40.6 metric tons)
Tare Weight:	abt. 8 Tonnes
Design Criteria:	DIN 15018 H2B4, nr. of lifting cycles 2 x 106
Telescoping System:	1 x 3,7kW
Expand/Retract from 20ft to 40ft position:	Approximately 28 seconds
Twistlock rotation:	90 degrees in abt. 1 second
Ambient Temperature:	-5 to +45 degrees celcius
Total power on spreader:	abt. 4,5kW
Number of rope sheaves:	8, outer diameter 707mm, KC type
Material of sheaves:	PA6 G
Hoisting rope dia:	18 mm

Electrical system

General requirements:	
Class of protection:	IP55
Standards and approvals:	EN, IEC
Power supply:	
Control voltage:	230 VAC, 50/60 Hz
Motor voltage:	460...480 VAC, 50/60 Hz
Spreader cable:	SPREADERFLEX YSLTOE-J36G2.5+6FO or equivalent

Junction box:
Material: Stainless steel
Spreader cable grommet: Splitting type, IP55 (Trelleborg MC 2 x 8-67 or equivalent)

Camera system

The spreader is fitted with four fixed focus cameras, one at each corner of spreader, looking down to twistlock area. Each camera is fitted to an adjustable bracket made of stainless steel. Cameras are connected to power source and fiber optic converter inside of the spreader's junction box.

The cameras have following particulars:

Camera type:	MC9150P, MOTEC
Operating voltage:	12 VDC
Connection type:	M12
Class of protection:	IP55
Fiber optic converter:	
Converter type:	FT400-SMT, OT SYSTEMS
Operating voltage:	12 VDC
Class of protection:	IP20
Power supply:	
Output voltage:	12 VDC
Output current:	2.5 A
Class of protection:	IP20

Cables from the junction box to the cameras need to be shielded, flexible and suitable for cable chain installation.

Cable type: M12-female connector at camera end, BNC-male + power supply wires at junction box end.

Surface protection

The spreader is prepared with high quality surface protection system designed for harsh industrial environment. Corrosive category is C5-I according to EN-ISO 12944. Durability class is M.

COMMERCIAL

Operation & Maintenance Manual

Two sets of fully detailed Operation and Maintenance Manuals will be supplied in English language. Each manual contains the spare parts list, operating instructions, maintenance instructions and general arrangement drawings of the spreader and mechanical systems.

Annex 7

Tech spec 8W Boxhunter

Annex 8

Tech spec 16W Boxhunter

Annex 9

Boxhunter - Supplier list

Annex 10

Training program

Mechanical training

TRAINING PROGRAMME
FOR RTG CRANES



Time	Day 1	Day 2	Day 3	Day 4	Day 5
AM 4 hours	KCI + VLC Introduction Introduction to the Crane Mechanical systems Mechanical drawing system Inspection schedules Classroom	Introduction to the Crane Brakes Rope change Bogie jacking Single turn Manual operation Crane			
PM 4 hours	Lubrication Brakes Couplings Gears Classroom	Bogie jacking Single turn Manual operation Crane			

Electrical training

KONECRANES®

Time	Day 1	Day 2	Day 3	Day 4	Day 5
AM 4 hours	Introduction to the Crane Crane Functions Crane Machinery	Drawing System Symbols and Abbreviations Use of Electrical Drawings	Drive system General Inverter hardware Monitoring values Parameter setting Manual operation	Drive system cont. PLC System overview & Operator's Interface Operator's information Maintenance information	Visit to Crane Trouble shooting
Lunch					
PM 4 hours	Introduction to the Crane Visit to Crane	GenSet Control Panel Equipment Panel Operations	Drive system cont. Visit to Crane Hardware of drives	PLC System overview & Operator's Interface cont. Operator's information Maintenance information	